



LINEARITY AND TIME-INVARIANCE

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Program:	BSCpE	Section:	A461

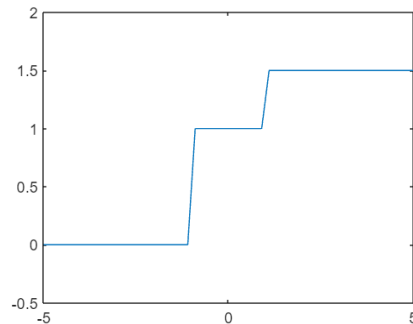
Example 1

Consider a system described by the I/O relationship $y(t) = tx(t)$.

Find out if this is a linear or a nonlinear system by using the signal with the following parameters:
 $a_1 = 1, x_1(t) = u(t + 1), a_2 = 0.5, x_2(t) = u(t - 1)$.

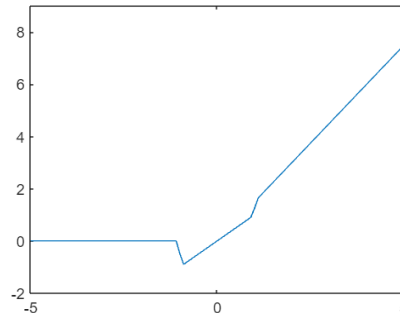
1. Plot the input signal $x(t) = u(t + 1) + 0.5u(t - 1)$:

```
t=-5:0.1:5;
xt1=heaviside(t+1);
xt2=0.5*heaviside(t-1);
xt=xt1+xt2;
plot(t, xt)
ylim([-0.5 2.0])
```



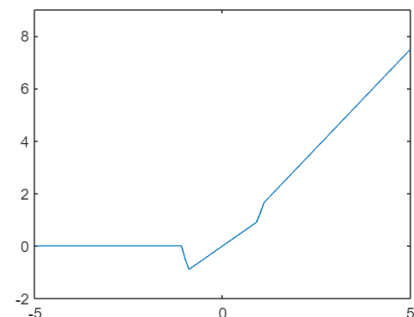
2. Plot the system $T[a_1x_1(t) + a_2x_2(t)]$:

```
y1=t.*xt;
plot(t, y1);
ylim([-2.0 9])
xlim([-5 5])
```



3. Then, plot the system $a_1T[x_1(t)] + a_2T[x_2(t)]$:

```
y2=t.*heaviside(t+1)+0.5*t.*heaviside(t-1);
plot(t, y2);
ylim([-2.0 9])
xlim([-5 5])
```



4. Since the plot of $T[a_1x_1(t) + a_2x_2(t)]$ is equal to $a_1T[x_1(t)] + a_2T[x_2(t)]$, therefore system $y(t) = tx(t)$ is **LINEAR**.



Problem Exercises

Determine whether the following systems are linear or nonlinear. Show your solution based on the given example and use **MATLAB** to verify its I/O relationship as linear or nonlinear systems.

For problems 1 – 3 use the following inputs to prove the linearity of the given systems using MATLAB: $a_1 = 0.2$, $x_1(n) = u(n)$, $a_2 = 0.4$, $x_2(n) = u(n - 10)$, $n = 0$ to 20.

1. $y(n) = 10\sin(0.1\pi)x(n)$
2. $y(n) = Ax(n) + B$; $A = 3, B = 2$
3. $y(n) = \frac{1}{4}x(n) + \frac{1}{2}x(n - 1) + \frac{1}{4}x(n - 2)$

For problems 4 – 6 use the following inputs to prove the linearity of the given systems using MATLAB: $a_1 = 0.8$, $x_1(t) = u(t - 5)$, $a_2 = 0.2$, $x_2(t) = t$, $t = 0$ to 15.

4. $y(t) = 3\cos(x(t))$
5. $y(t) = 3\cos(t)x(t)$
6. $y(t) = e^{x(t)}$



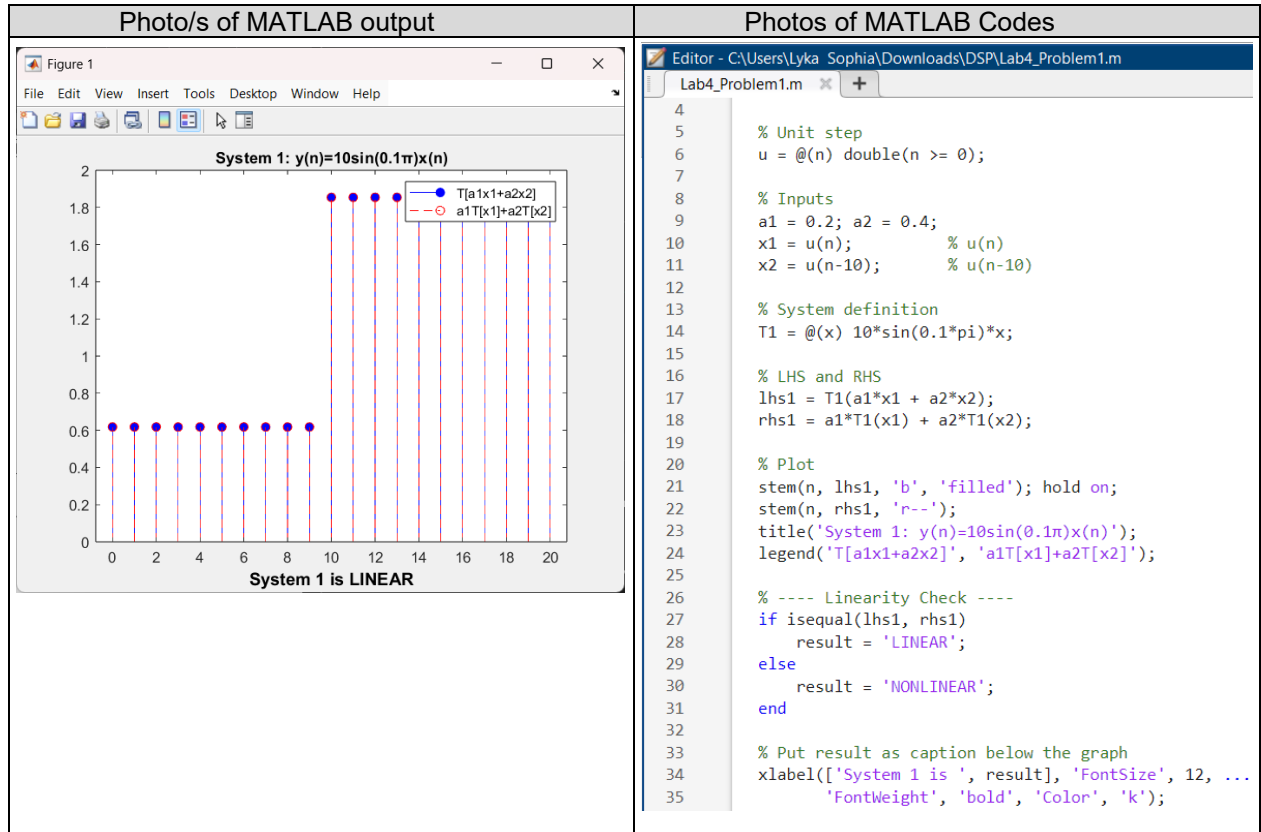
LINEARITY AND TIME-INVARIANCE

A. Problem Exercises:

(Please include the following: 1. Photos of the codes and 2. Outputs after executing the MATLAB code.)

Determine the relationship of the given systems as linear or nonlinear in the Problem Exercises.

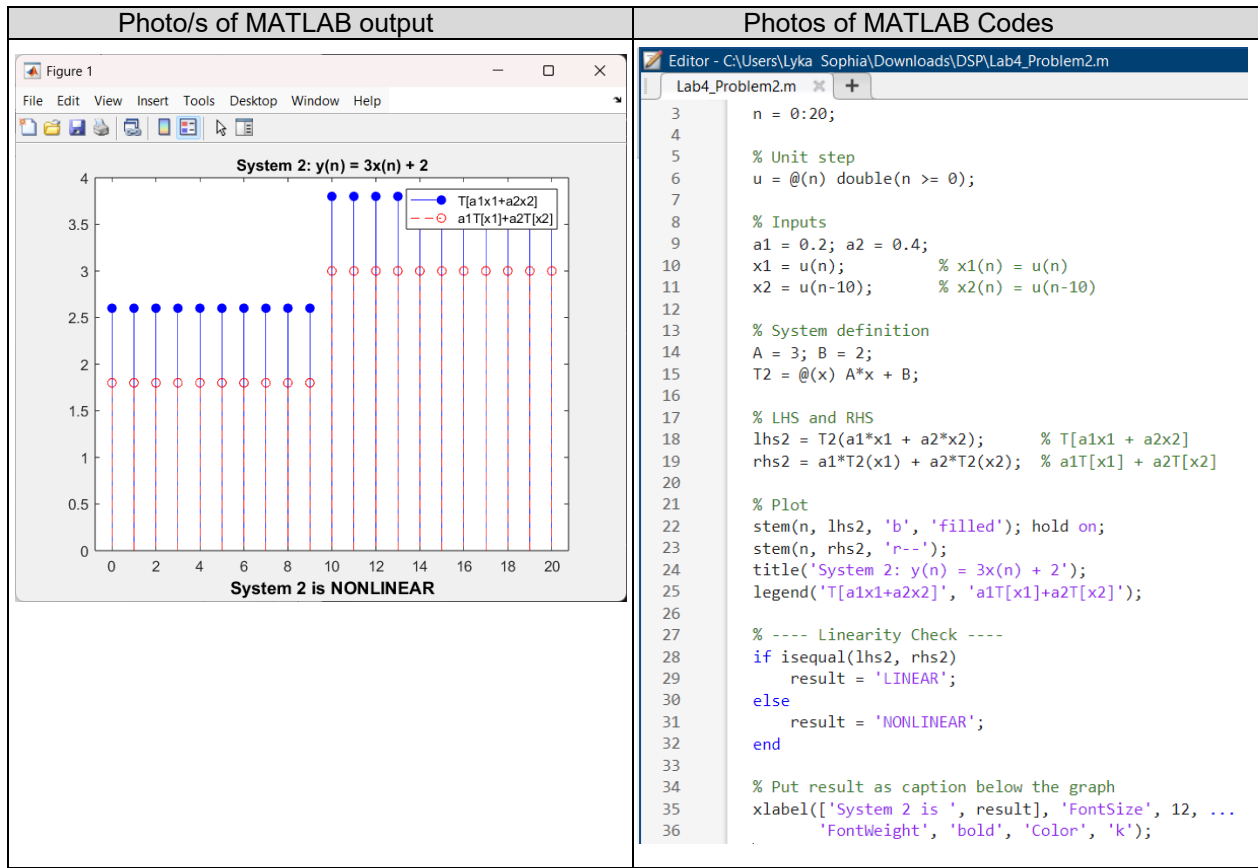
Problem 1:





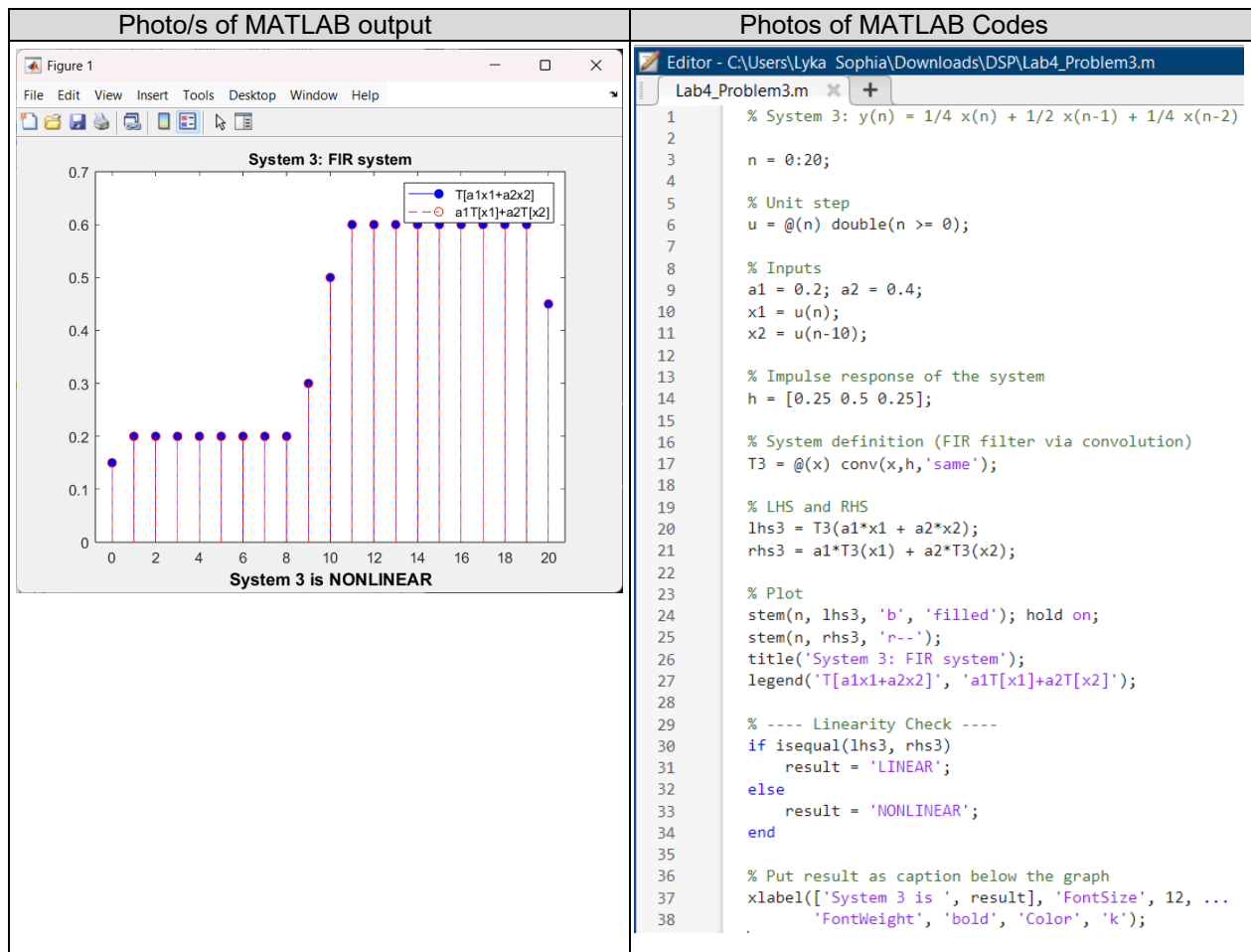
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Problem 2:



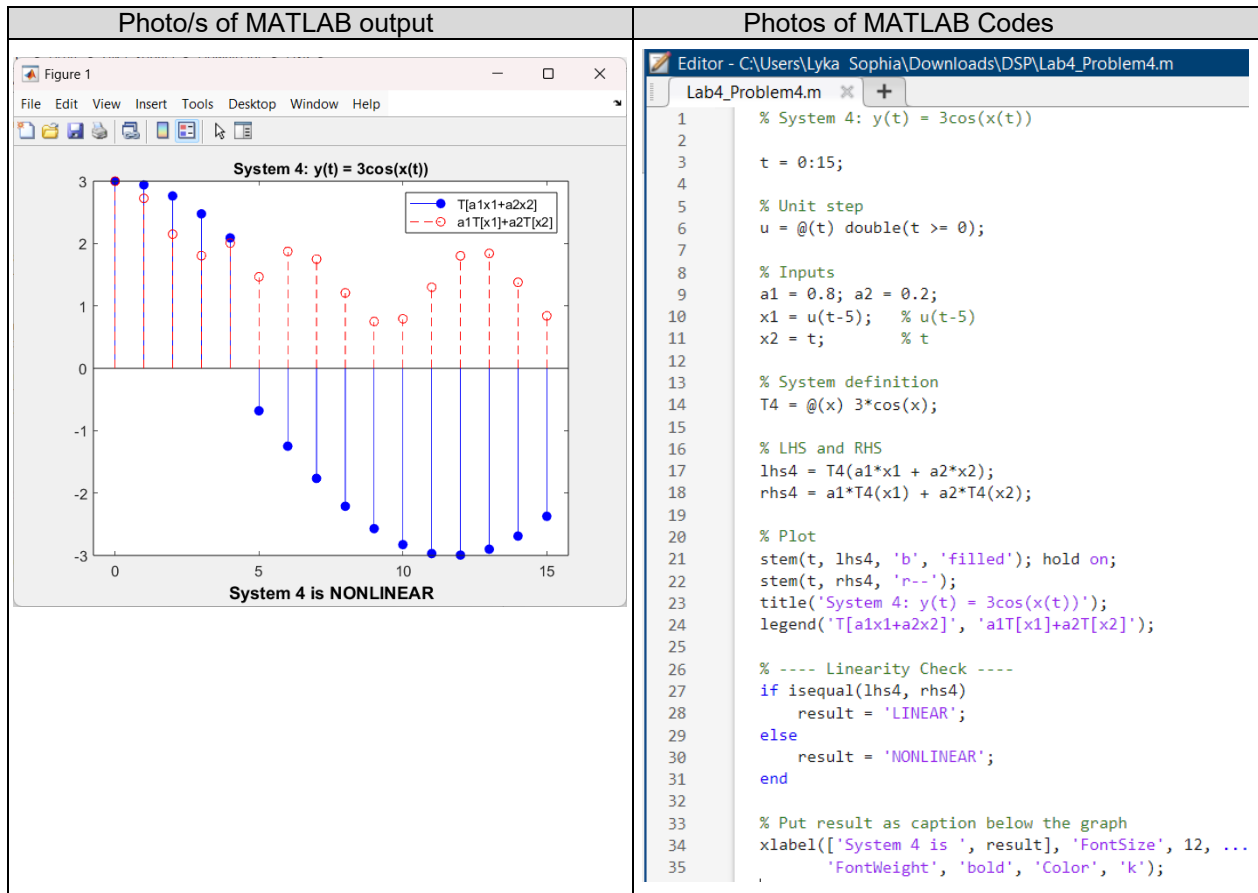
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Problem 3:



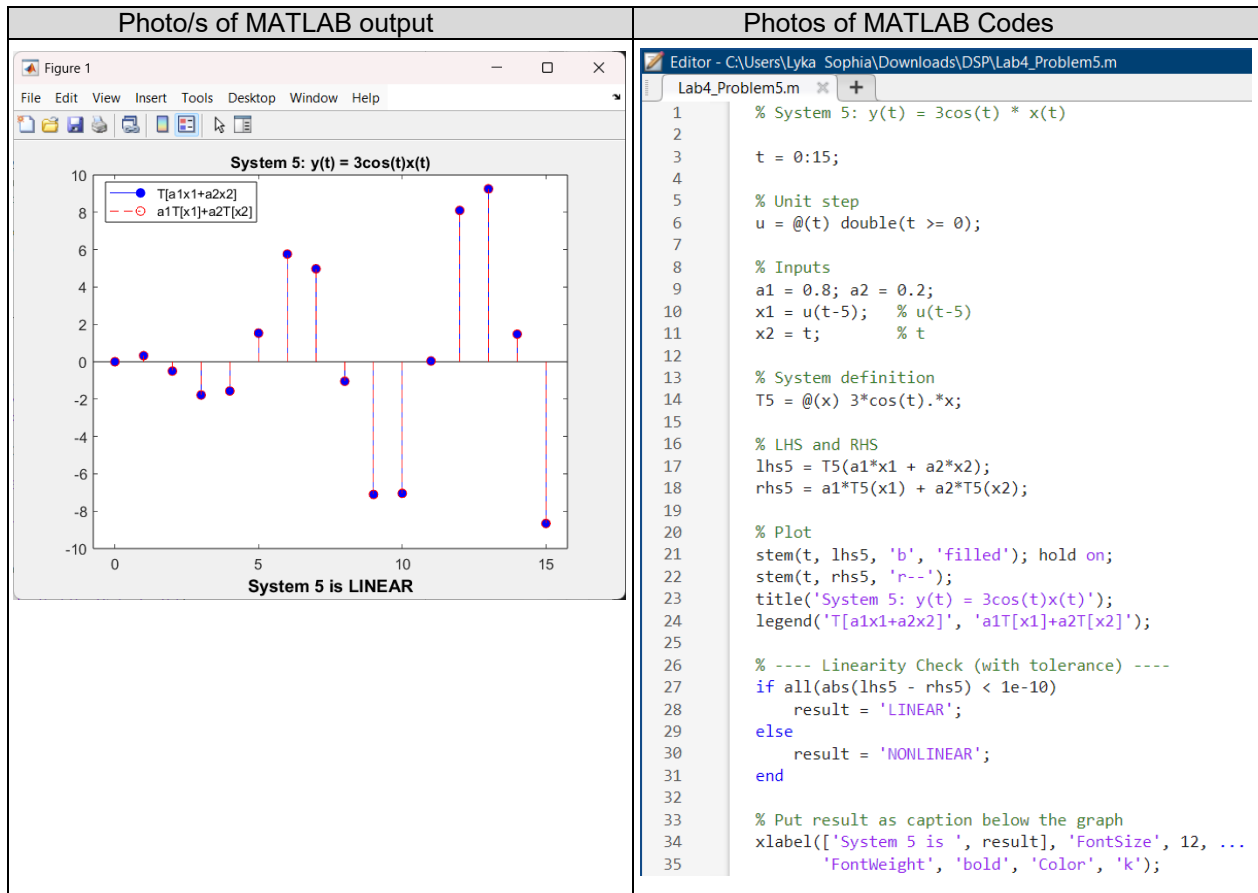
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Problem 4:

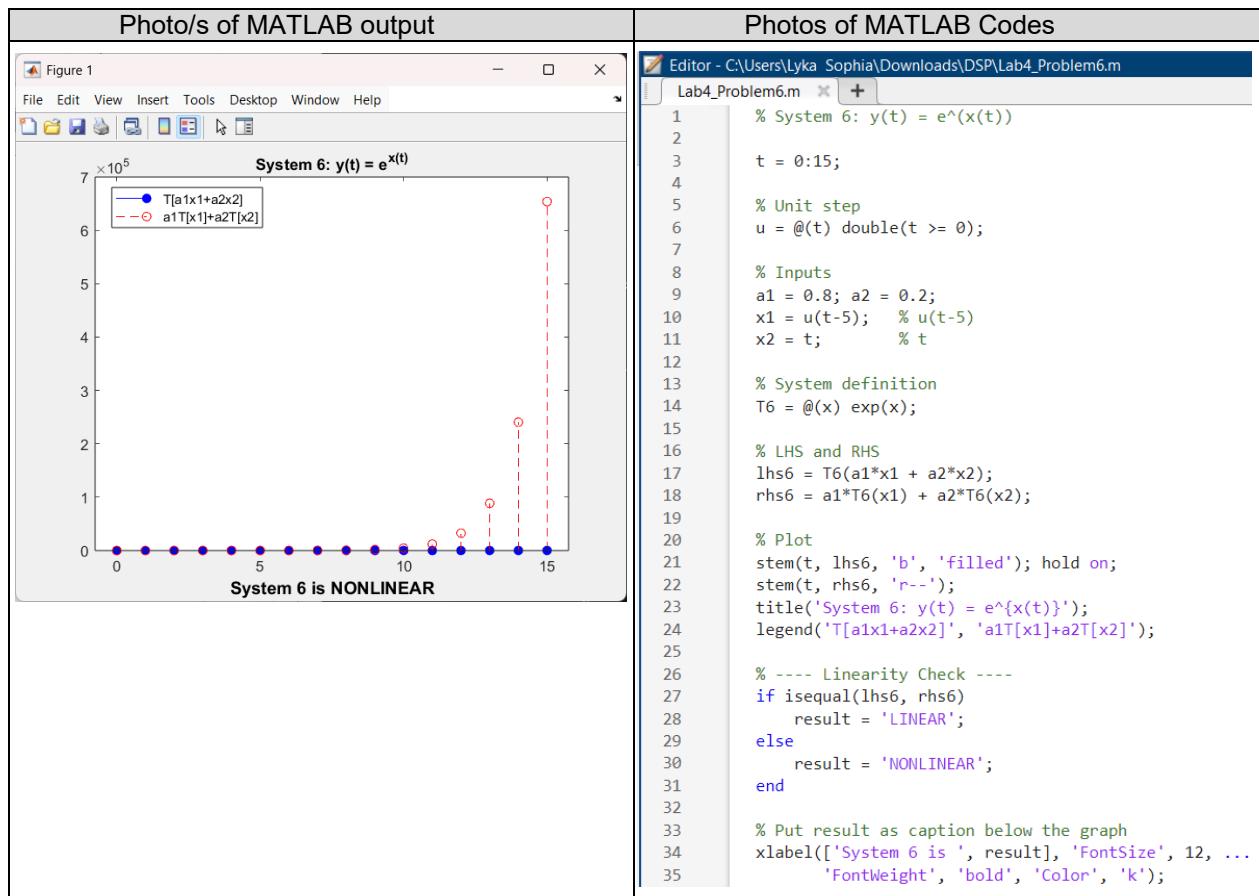


LINEARITY AND TIME-INVARIANCE

Problem 5:



Problem 6:



B. Findings ,Observations, and Comments:

Guiding Questions :

1. What did you learn in the guided exercises? Did you learn something new?
2. What are your findings, observations, and comments in the guided exercises?
3. What are your findings, observations, and comments in the problem exercises?

From the guided exercises, I learned how to check if a system is linear or nonlinear. I also learned how to use MATLAB to test the superposition property and confirm the results. This gave me a clearer idea of how theory and simulation connect, and I picked up new ways of checking my answers more confidently.

My findings and observations from the guided exercises are that linear systems are easier to analyze because their outputs combine directly, while nonlinear systems break this rule. I also noticed that constants or nonlinear functions like cosine and exponential usually make the system nonlinear. Overall, MATLAB made it easier to see the differences through graphs and results.

In the problem exercises, my findings are that some systems like 1, 3, and 5 are linear, while others like 2, 4, and 6 are nonlinear. My observation is that even when the graphs sometimes look the same, the system can still be nonlinear depending on its equation. My main comment is that the exercises were helpful in showing how to carefully analyze systems and not just rely on appearance.